



University of
Nottingham

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COMP4139

Machine Learning (MLE)

Sequence Models (RNN & LSTM)

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Learning Objectives

- Recurrent Neural Network (RNN)
 - Working principles
 - What it might suffer from
- Long Short-Term Memory (LSTM)
 - Working principles
 - Why could it be a “better” choice
- Application examples (i.e., text and image captioning)

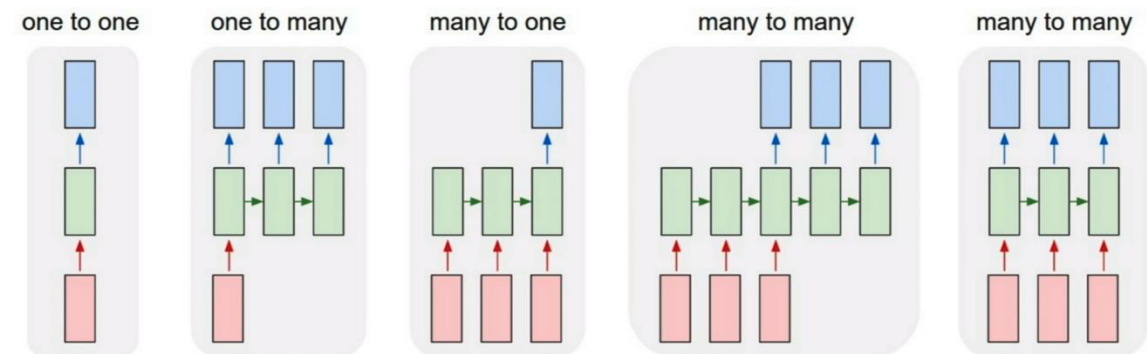


Sequential Data Learning

- One data item is dependent on those that come before or after it (**not** independently and identically distributed (i.i.d.)
- Machine learning models that input or output data sequences are known as sequence models.
- Data
 - Text streams
 - audio clips
 - video clips
 - time-series data (weather, stocks, etc.)
- Applications
 - Speech/voice recognition
 - Weather forecast
 - Language translation
 - DNA Sequence Analysis

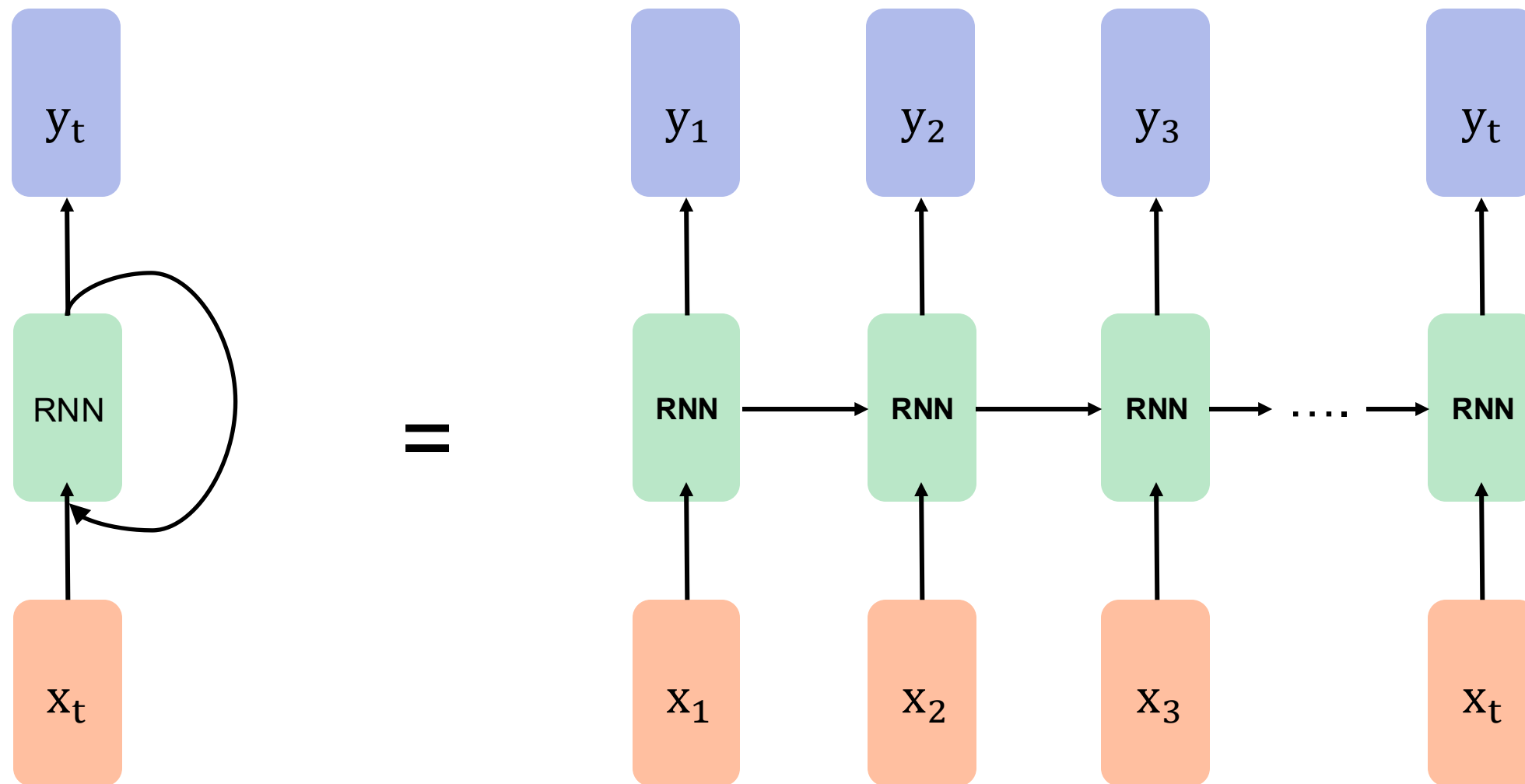
Application Scenarios of Recurrent Neural Networks

- One-to-one
 - Classical feed forward neural network with one input and one output. (e.g. image classification)
- One-to-many
 - Input image and output words with variable length (e.g. image captioning, music generation)
- Many-to-one
 - E.g. sentiment classification; share price prediction
- Many-to-many
 - Language translation: input and output are with variable length.
 - Video clip classification: input and output have the same length.



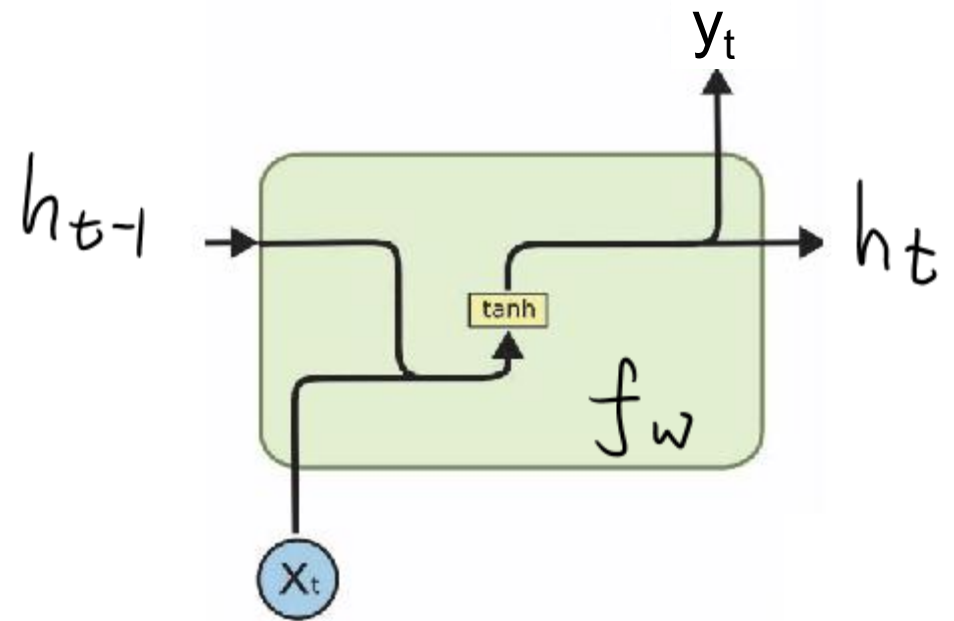


RNN rolled = unrolled





Basic RNN Unit



Basic RNN Unit

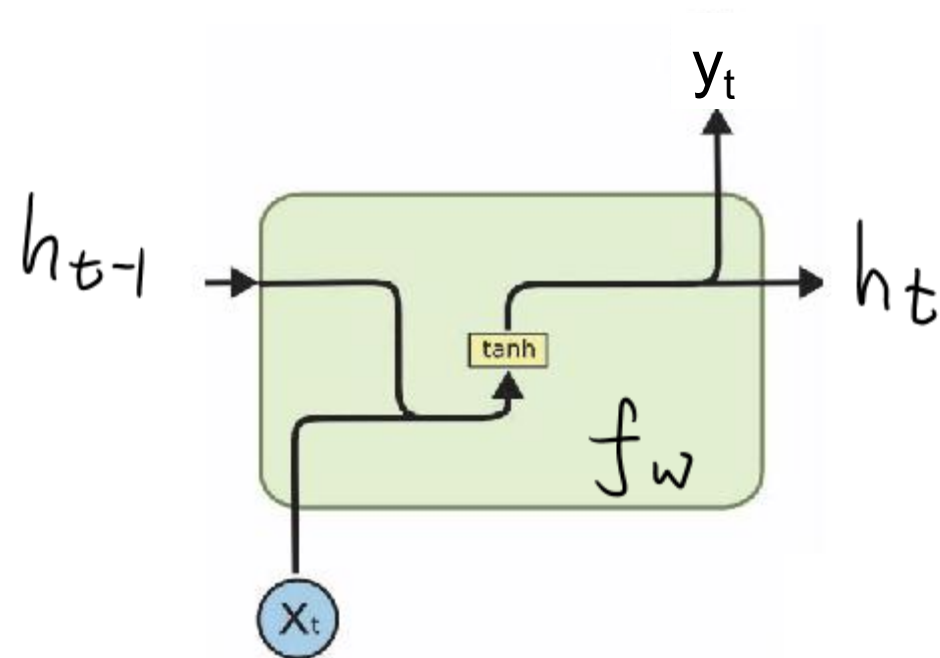
Current state Previous state

$$h_t = f_w(h_{t-1}, x_t)$$

Mapping/activation
function Input vector at time step t

$$h_t = \tanh(w_{hh}h_{t-1} + w_{xh}x_t + b_{ht})$$

$$y_t = W_{hy}h_t + b_{yt}$$



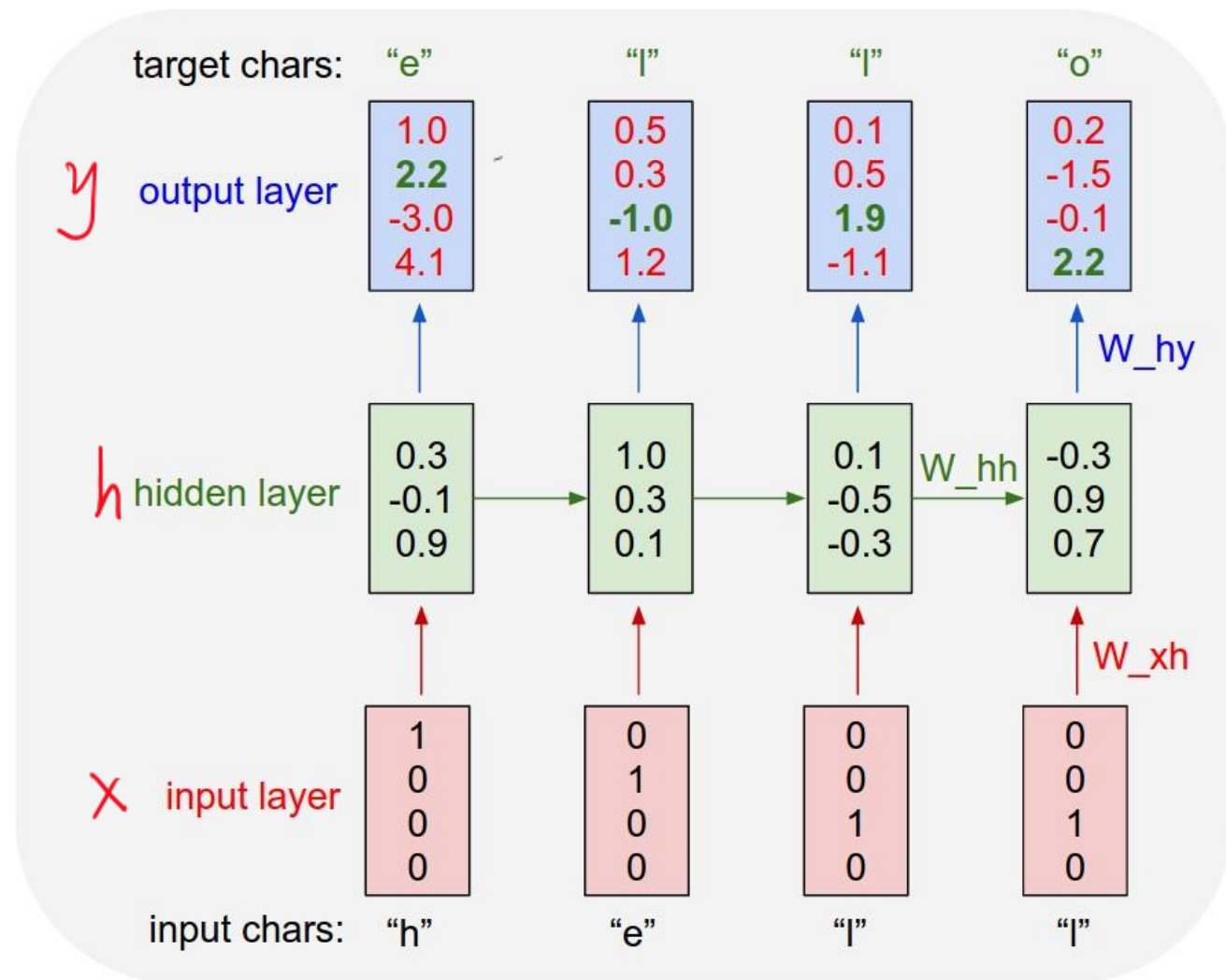
x is the input vector
 h is a hidden vector
 y is the output vector

A Simple Recurrent Neural Network

- Example of character-level language model
- Training example: “hello”
- Vocabulary: [h, e, l, o]
- The **same** W_{hh} , W_{hy} and W_{xh} are used for all RNN units

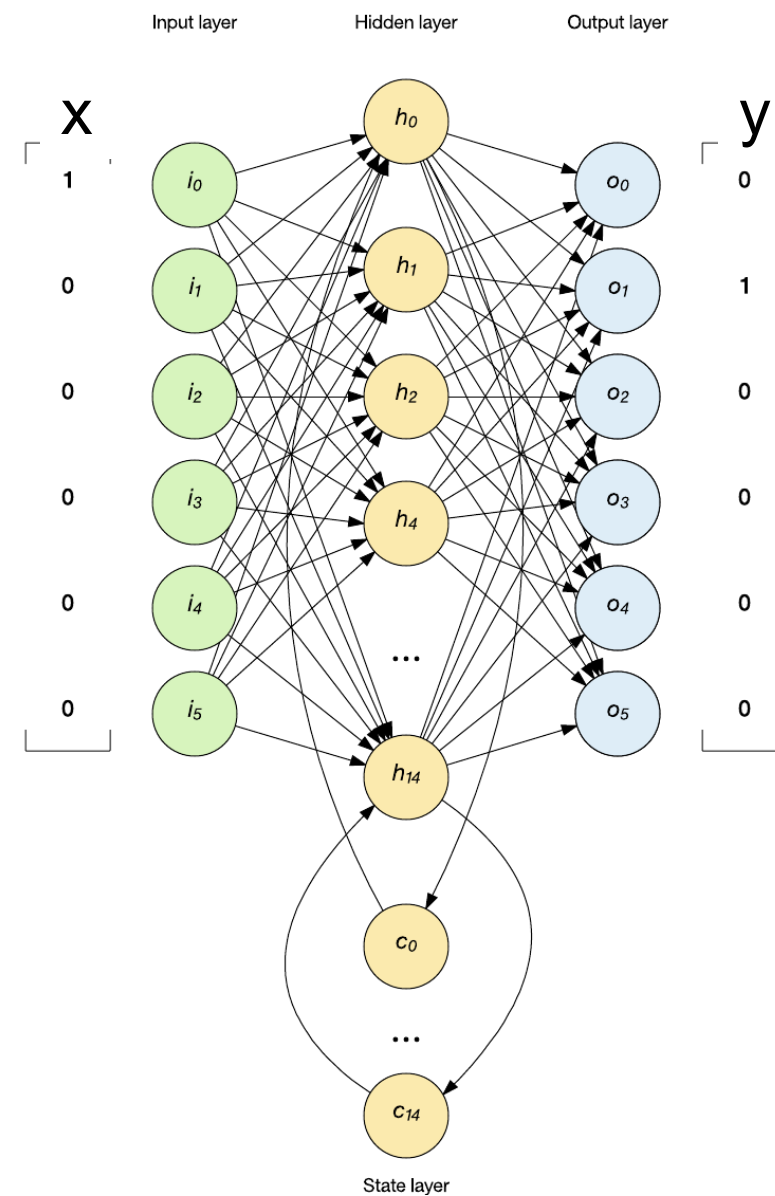
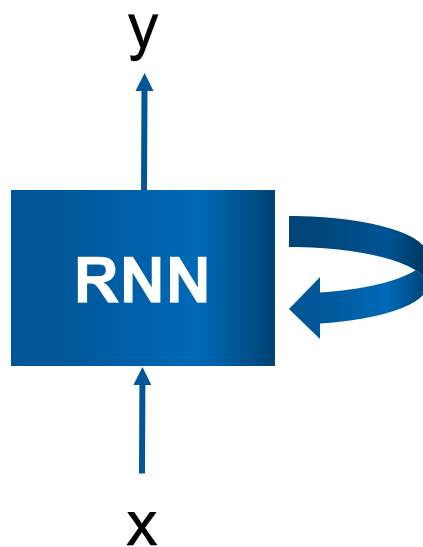
$$h_t = \tanh(w_{hh}h_{t-1} + w_{xh}x_t + b_{ht})$$

$$y_t = W_{hy}h_t + b_{yt}$$





RNN Structure



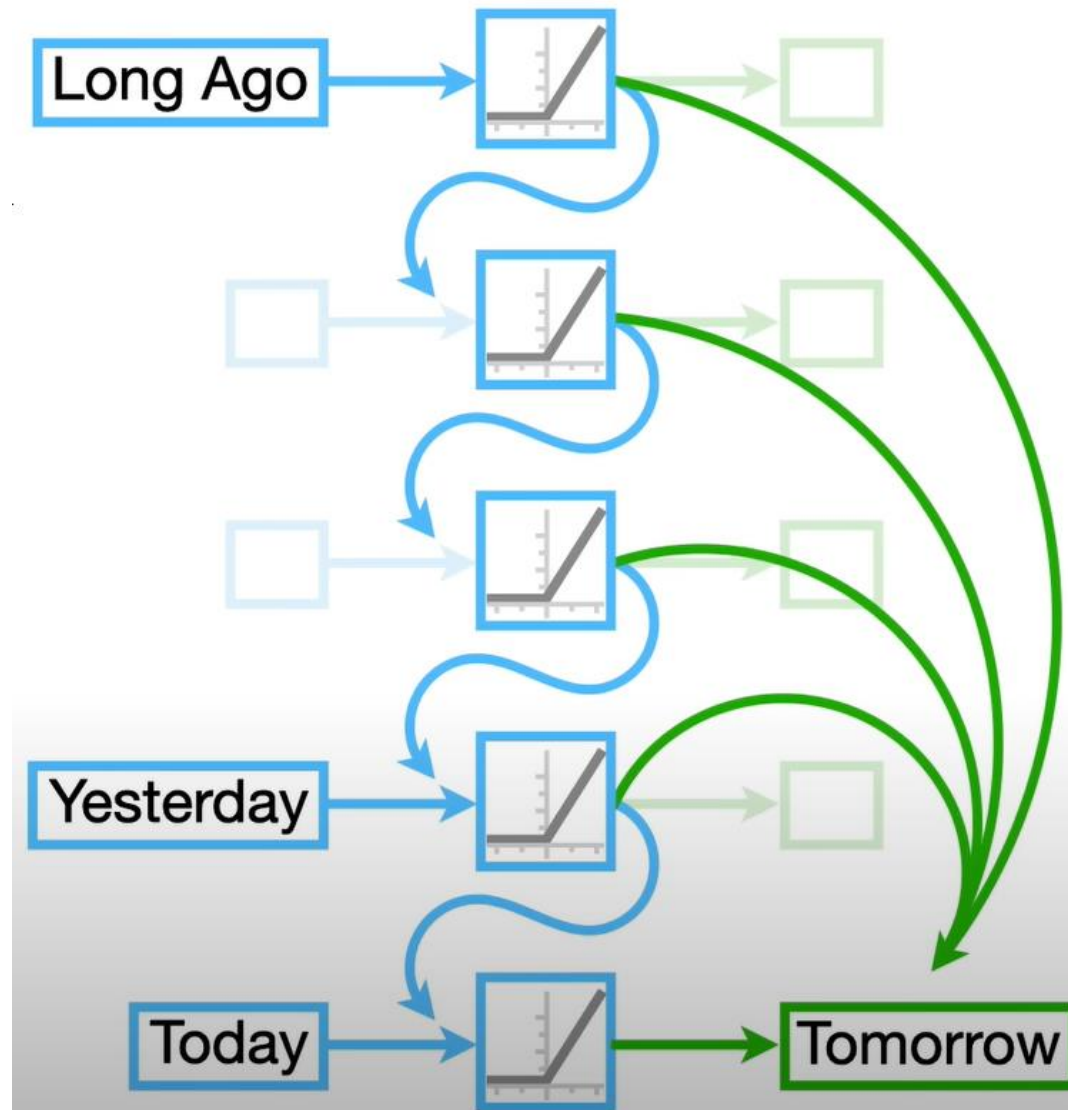


Long Short Term Memory Networks (LSTM)

- Vanilla RNN suffers from **vanishing/exploding** gradient problem.
- LSTM was introduced by Schmidhuber and Hochreiter in 1997.
- It is explicitly designed to capture long term dependency.
- Remembering the long sequences for a long period of time.
- Besides **hidden state h** , LSTM also has a **cell state c** to remember things from previous knowledge.
- LSTM back propagate gradient much more efficiently than vanilla RNN



LSTM General Idea

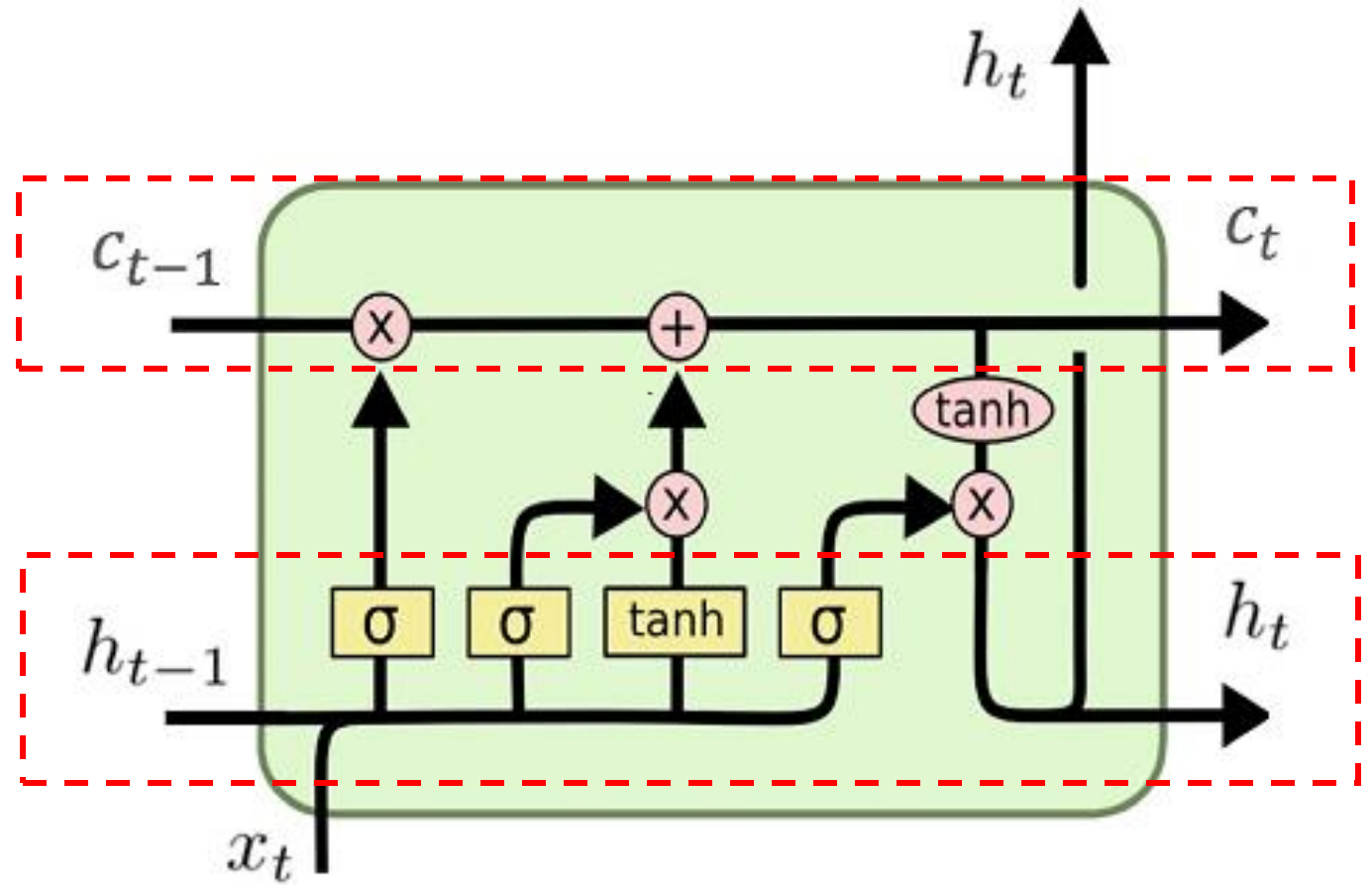


LSTM General Idea

Long term memory:
cell states (non-learnable)

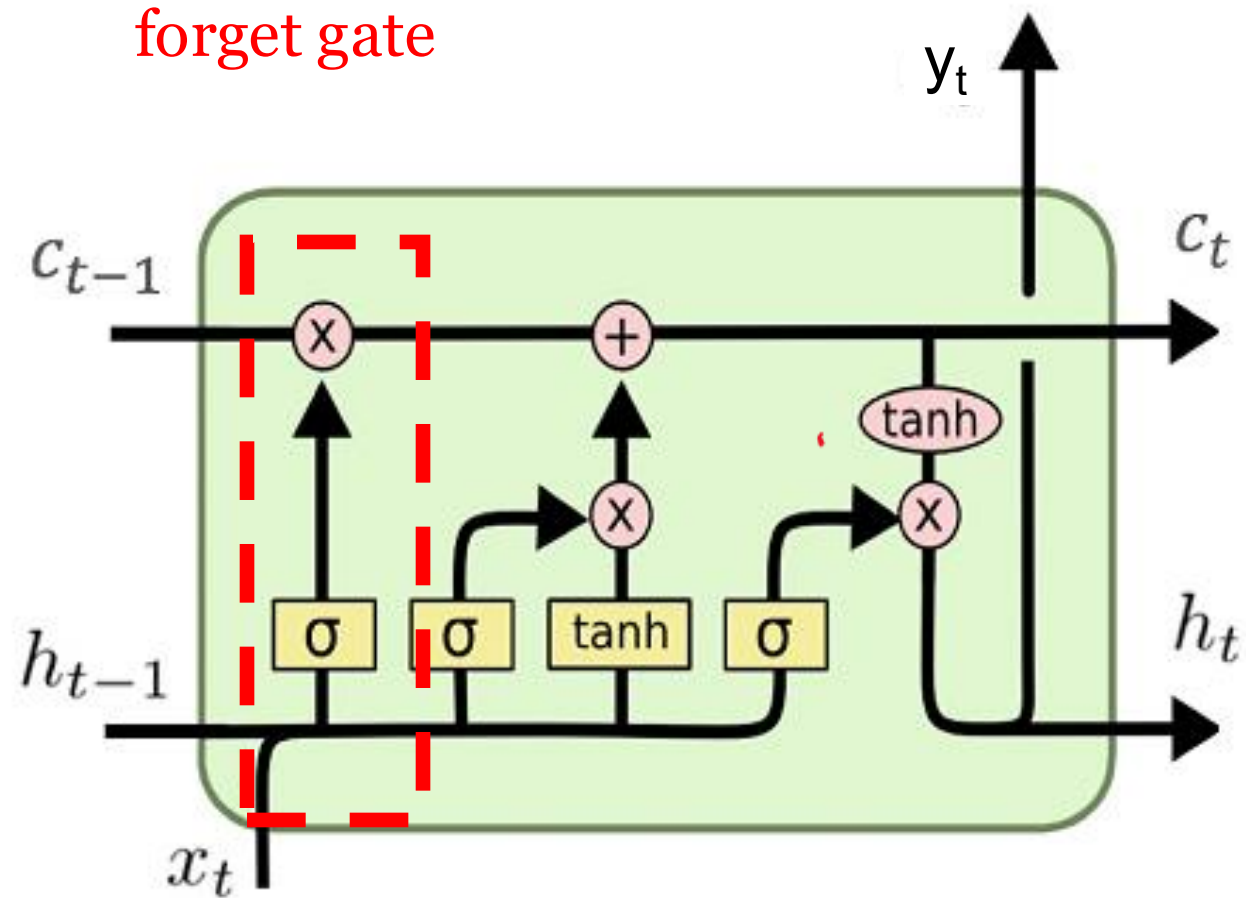
Short term memory:
hidden states (learnable)

Sigmoid functions act as gates to
switch on/off the information passing



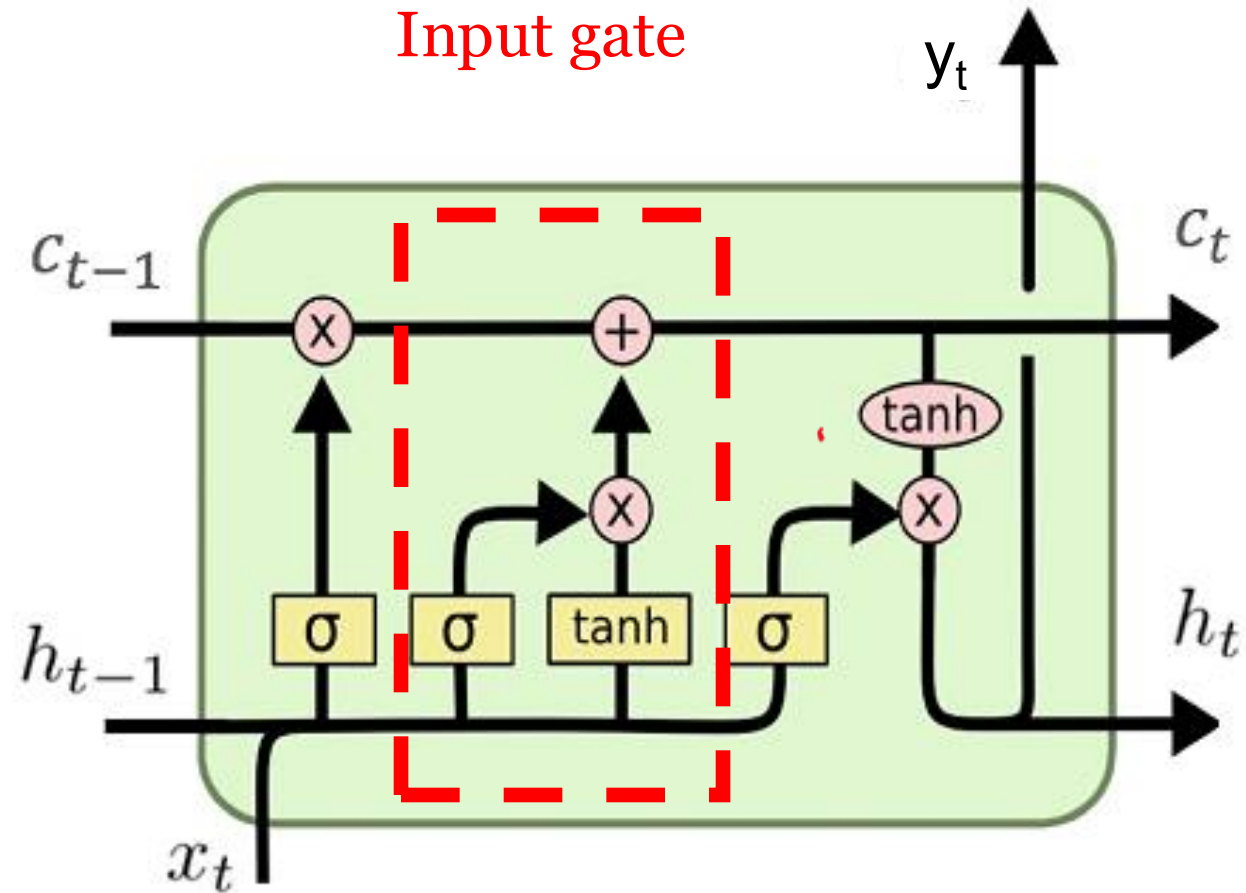
LSTM
(Long-Short Term Memory)

Forget Gate



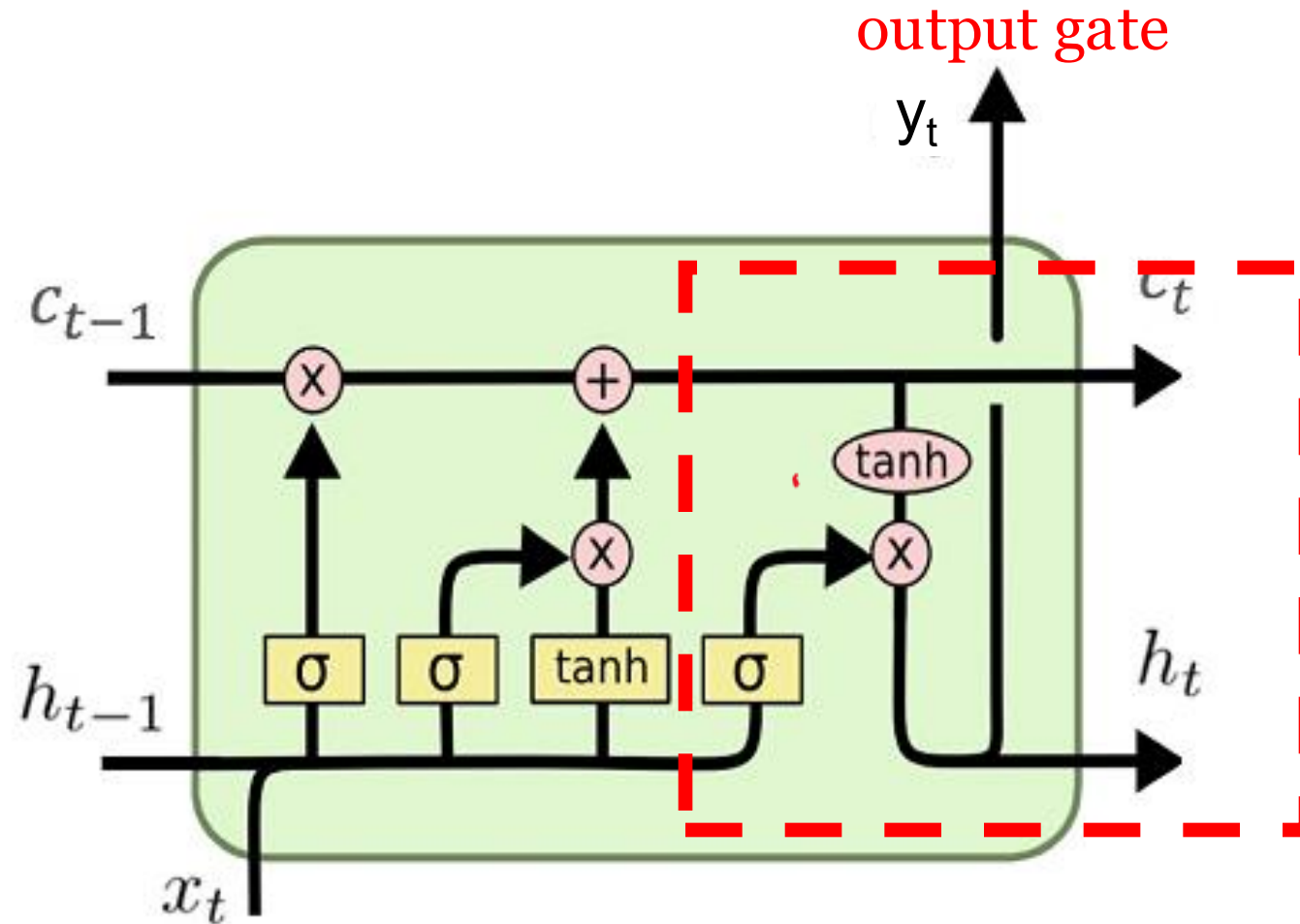
LSTM
(Long-Short Term Memory)

Input Gate



LSTM
(Long-Short Term Memory)

Output Gate



LSTM
(Long-Short Term Memory)

LSTM Unit- Forget Gate

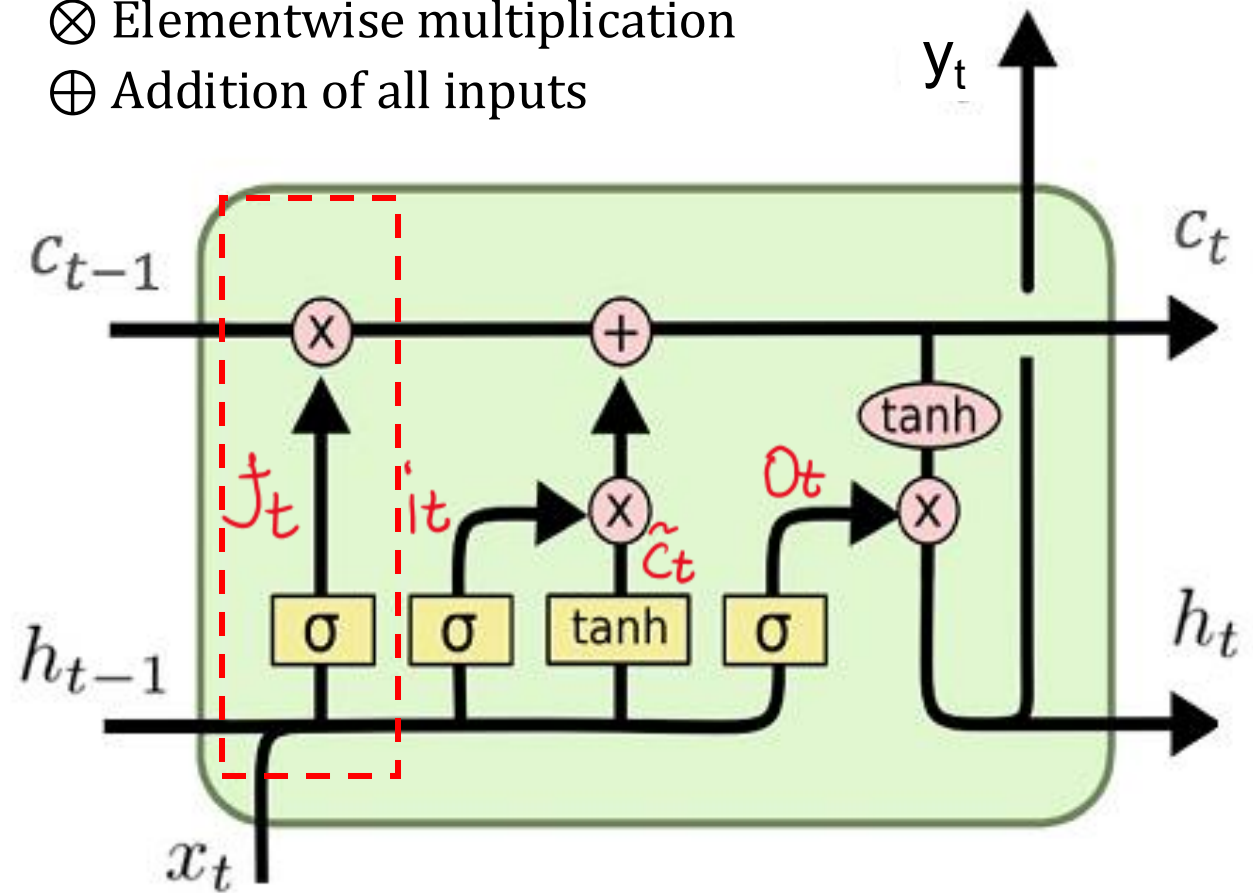
f_t : forget gate

$$f_t = \text{sigmoid}(w_f h_{t-1} + u_f x_t + b_f)$$

The forget gate determines how much (%) long term memory to retain.

⊗ Elementwise multiplication

⊕ Addition of all inputs



LSTM
(Long-Short Term Memory)

LSTM Unit- Input Gate

i_t : input gate

$$i_t = \text{sigmoid}(w_i h_{t-1} + u_i x_t + b_i)$$

$\tilde{c}_t$: memory cell candidate

$$\tilde{c}_t = \tanh(w_c h_{t-1} + u_c x_t + b_c)$$

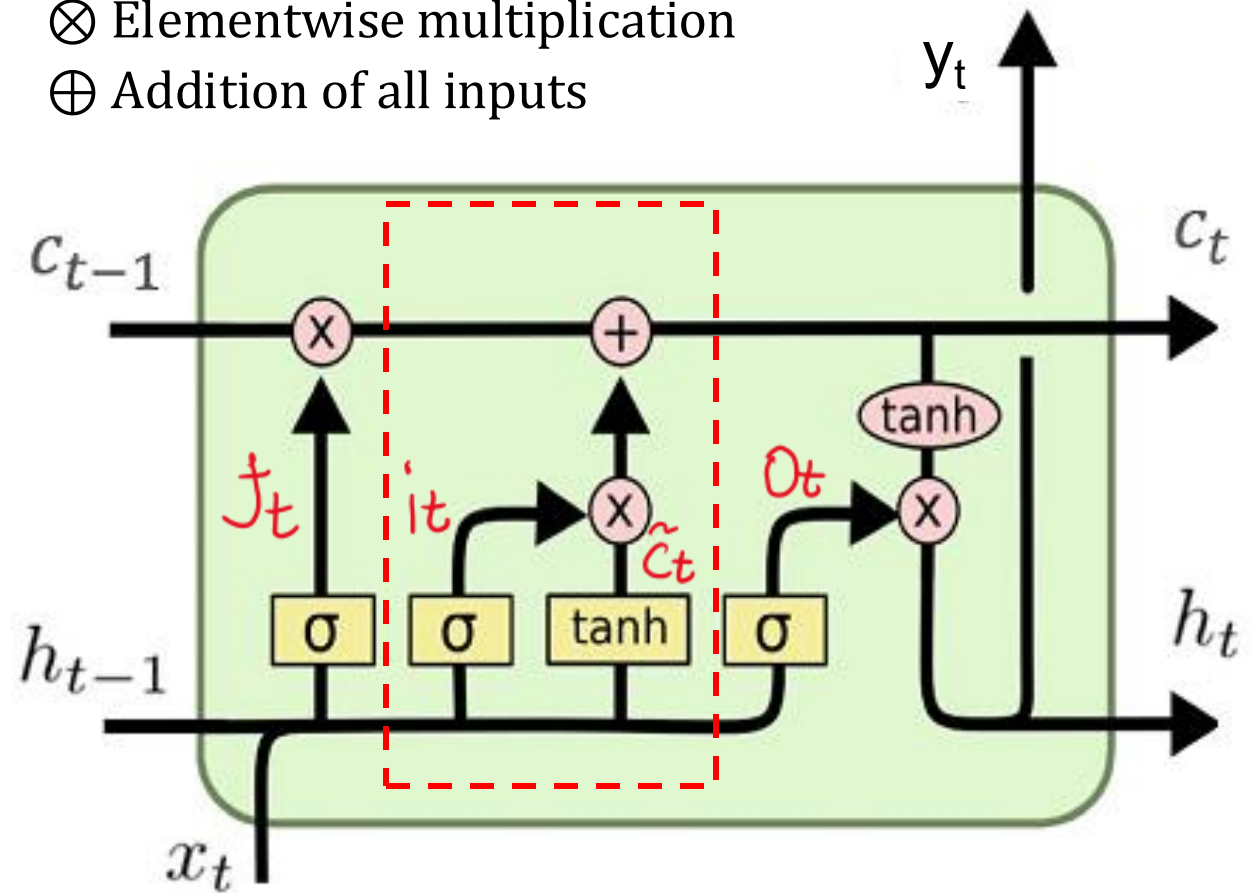
c_t : memory cell

$$c_t = f_t \otimes c_{t-1} + i_t \otimes \tilde{c}_t$$

The input gate determines how much (%) of the short-term information should be contributed to the long-term memory C_t .

$\otimes$ Elementwise multiplication

$\oplus$ Addition of all inputs



LSTM
(Long-Short Term Memory)

LSTM Unit- Output Gate

o_t : output gate

$$o_t = \text{sigmoid}(w_o h_{t-1} + u_o x_t + b_o)$$

h_t : hidden state

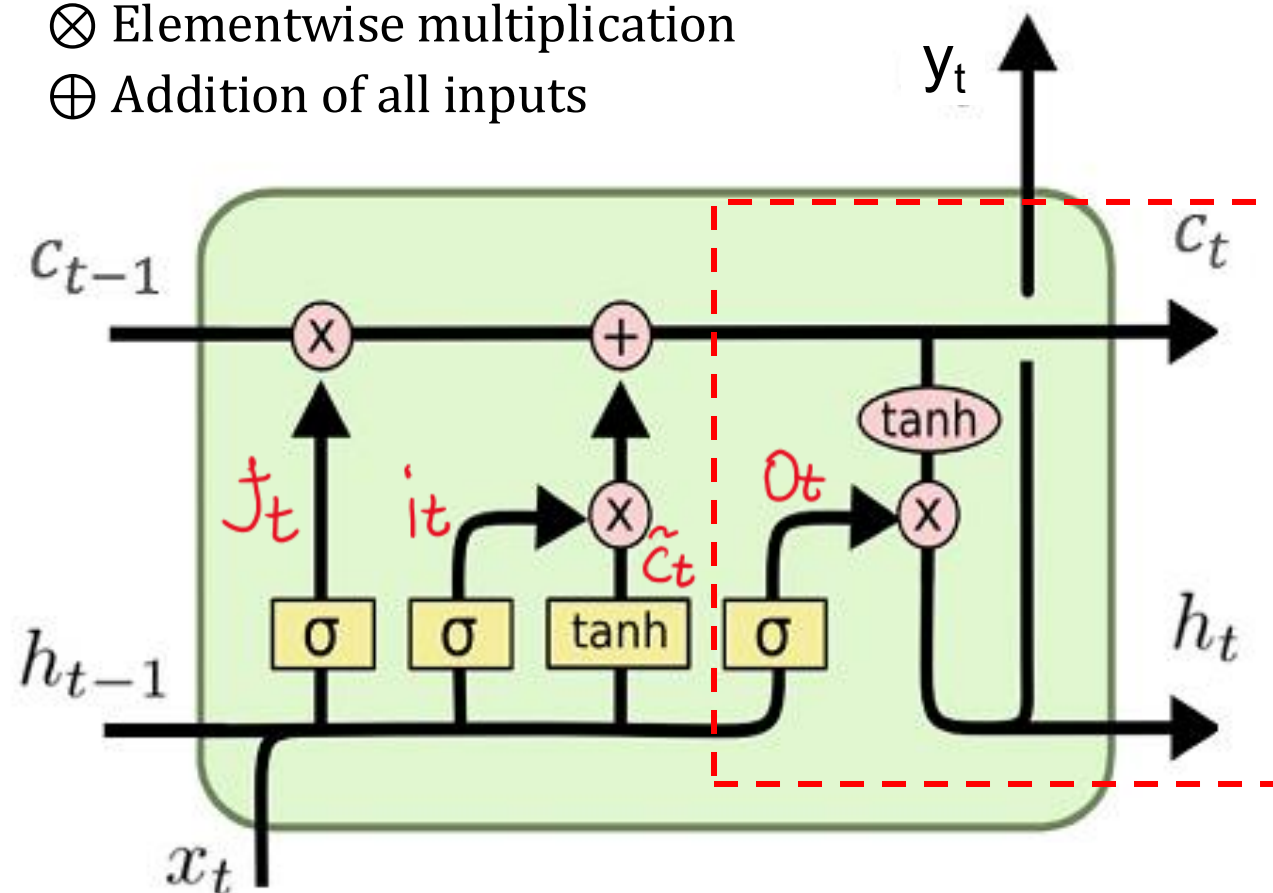
$$h_t = o_t \otimes \tanh(c_t)$$

$$y_t = h_t$$

The output gate determines how much (%) of the new long-term memory should be contributed to the current output $y_t = h_t$.

$\otimes$ Elementwise multiplication

$\oplus$ Addition of all inputs



LSTM
(Long-Short Term Memory)

LSTM Unit- All together

f_t : forget gate

$$f_t = \text{sigmoid}(w_f h_{t-1} + u_f x_t + b_f)$$

i_t : input gate

$$i_t = \text{sigmoid}(w_i h_{t-1} + u_i x_t + b_i)$$

$\tilde{c}_t$: memory cell candidate

$$\tilde{c}_t = \tanh(w_c h_{t-1} + u_c x_t + b_c)$$

c_t : memory cell

$$c_t = f_t \otimes c_{t-1} \oplus i_t \otimes \tilde{c}_t$$

o_t : output gate

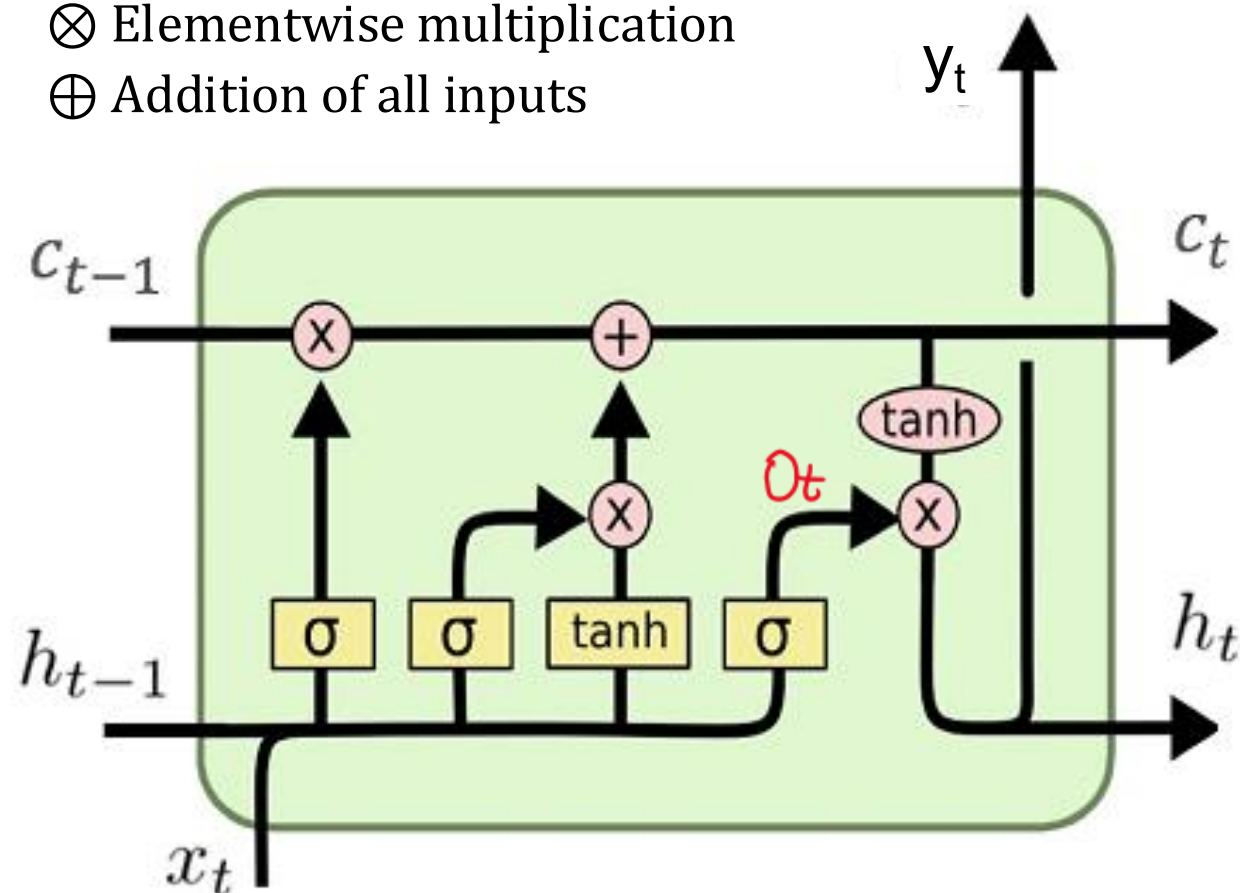
$$o_t = \text{sigmoid}(w_o h_{t-1} + u_o x_t + b_o)$$

h_t : hidden state

$$h_t = o_t \otimes \tanh(c_t) \quad y_t = h_t$$

$\otimes$ Elementwise multiplication

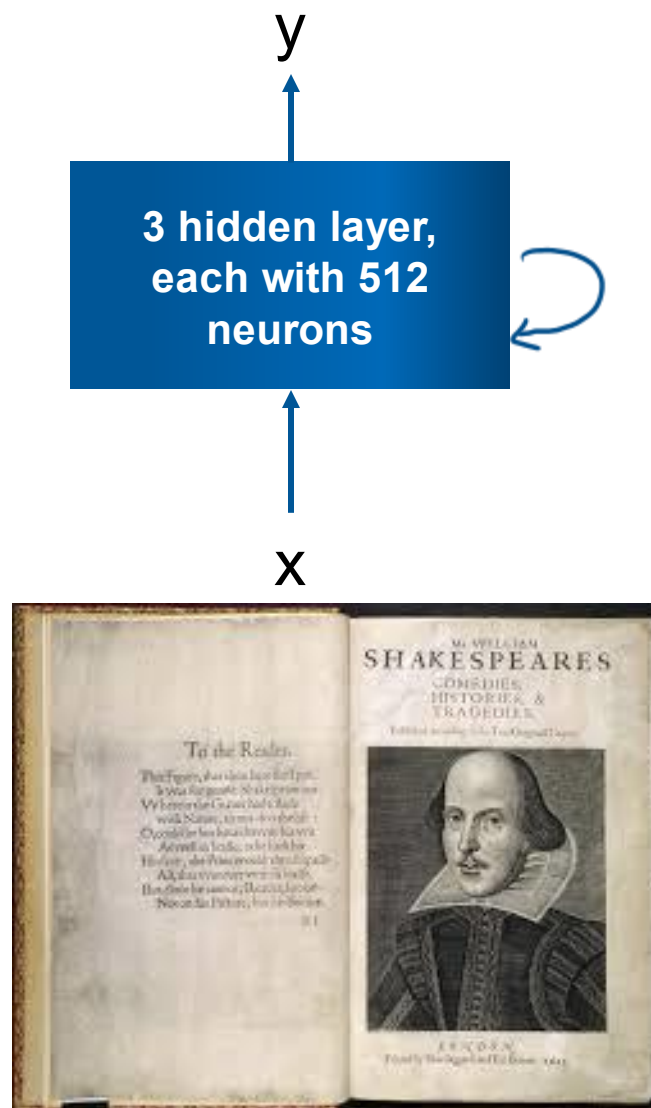
$\oplus$ Addition of all inputs



LSTM
(Long-Short Term Memory)



Applications: Text Generation



Outputs after training

PANDARUS:

Alas, I think he shall be come approached and the day
When little strain would be attain'd into being never fed,
And who is but a chain and subjects of his death,
I should not sleep.

Second Senator:

They are away this miseries, produced upon my soul,
Breaking and strongly should be buried, when I perish
The earth and thoughts of many states.

DUKE VINCENTIO:

Well, your wit is in the care of side and that.

Second Lord:

They would be ruled after this chamber, and
my fair nues begun out of the fact, to be conveyed,
Whose noble souls I'll have the heart of the wars.

Clown:

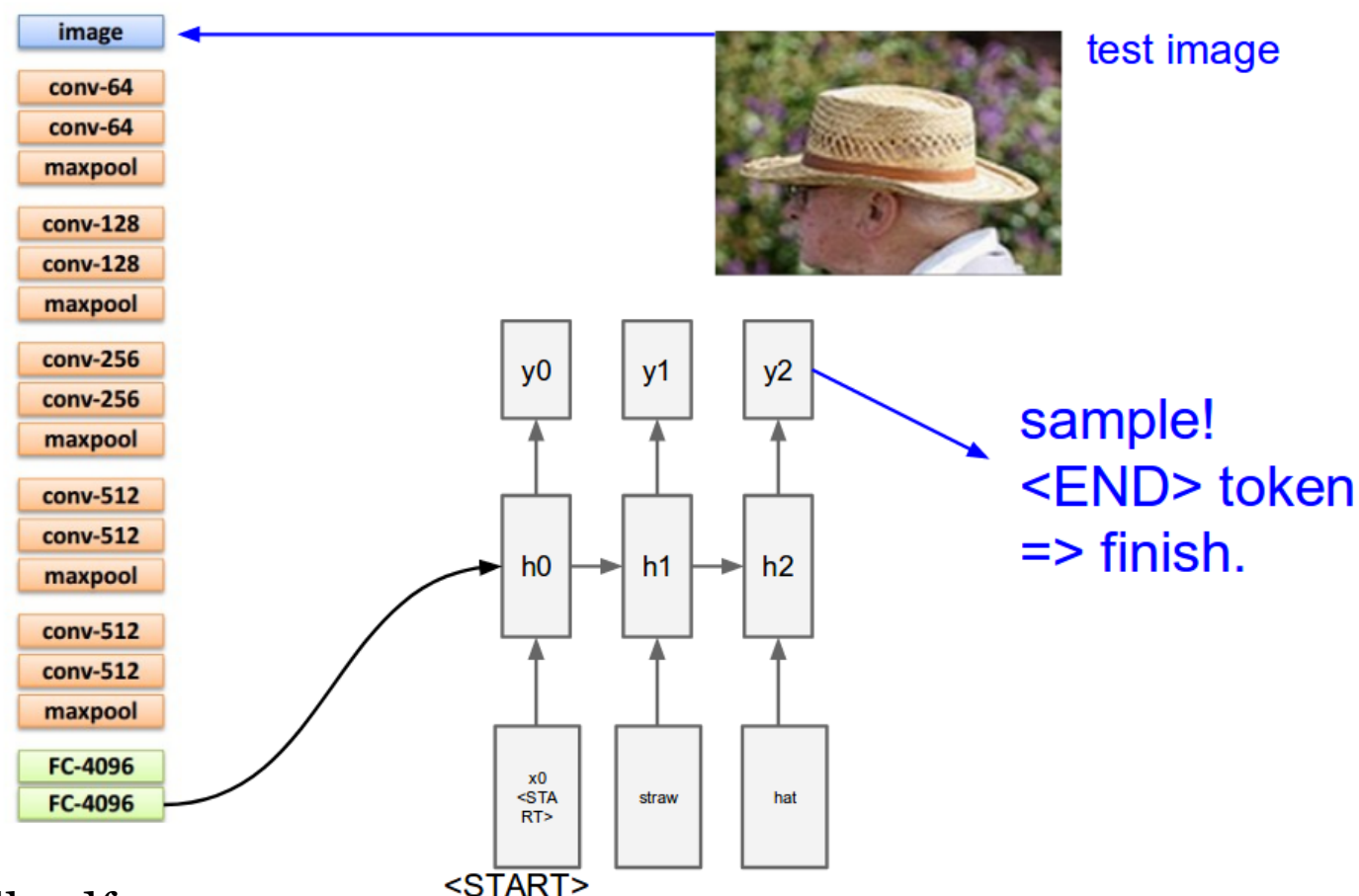
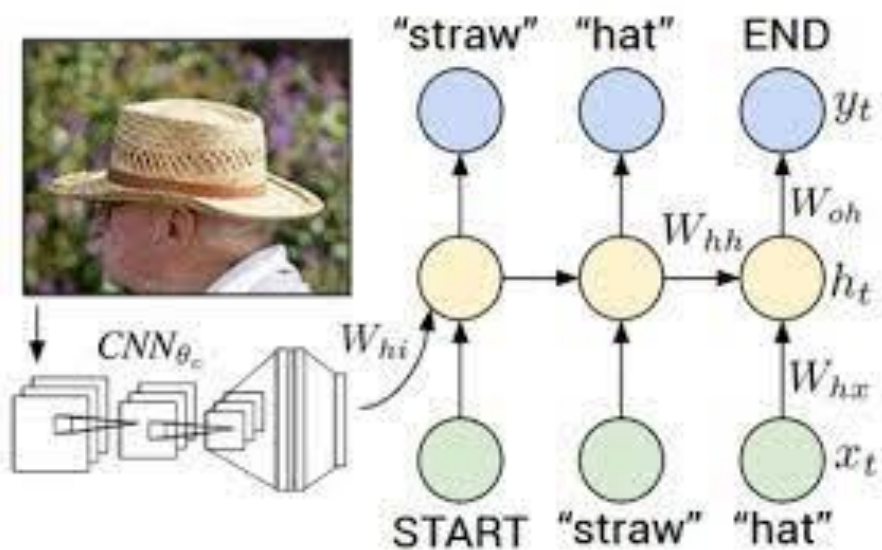
Come, sir, I will make did behold your worship.

VIOLA:

I'll drink it.

Applications: Image Captioning

- Pre-trained CNN classification model (e.g. VGG)
- Feed the learned feature vector to the first hidden layer
- One hot encoding of a 300-word vocabulary
- End to end training of RNN



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